10. Develop a Course Registration and Academic Management System using Spring MVC, Hibernate ORM, JPA, Dependency Injection, Annotations, and MySQL Database Integration.

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Course Registration</title>
</head>
<body>
  <h2>Register New Course</h2>

  <form action="save-course" method="post">
    <label>Course Code:</label><br>
    <input type="text" name="courseCode" placeholder="e.g. CS101" required><br><br>

    <label>Course Title:</label><br>
    <input type="text" name="title" placeholder="e.g. Data Structures" required><br><br>

    <label>Credits:</label><br>
    <input type="number" name="credits" min="1" max="6" required><br><br>

    <button type="submit">Register Course</button>
  </form>
</body>
</html>
```

Course.java

```
package com.example.course;

import jakarta.persistence.*;
```

```
@Entity
@Table(name = "courses")
public class Course {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private int id;

    private String courseCode;
    private String title;
    private int credits;

    public Course() {}

    public Course(String courseCode, String title, int credits) {
        this.courseCode = courseCode;
        this.title = title;
        this.credits = credits;
    }

    // Getters and Setters
    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getCourseCode() { return courseCode; }
    public void setCourseCode(String courseCode) { this.courseCode = courseCode; }

    public String getTitle() { return title; }
```

```
    public void setTitle(String title) { this.title = title; }

    public int getCredits() { return credits; }
    public void setCredits(int credits) { this.credits = credits; }
}
```

CourseDao.java

```
package com.example.course;

import jakarta.persistence.EntityManager;
import jakarta.persistence.PersistenceContext;
import org.springframework.stereotype.Repository;
import org.springframework.transaction.annotation.Transactional;
import java.util.List;

@Repository
public class CourseDao {

    @PersistenceContext
    private EntityManager em;

    @Transactional
    public void save(Course course) {
        em.persist(course);
    }

    public List<Course> getAll() {
        return em.createQuery("SELECT c FROM Course c", Course.class).getResultList();
    }
}
```

CourseController.java

```
package com.example.course;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.*;
import java.util.List;

@Controller
public class CourseController {

    @Autowired
    private CourseDao courseDao;

    @PostMapping("/save-course")
    public String registerCourse(@ModelAttribute Course course) {
        courseDao.save(course);
        return "redirect:/courses";
    }

    @GetMapping("/courses")
    @ResponseBody
    public List<Course> getAllCourses() {
        return courseDao.getAll();
    }
}
```

AppConfig.java

```
package com.example.course;
```

```
import org.springframework.context.annotation.*;
import org.springframework.jdbc.datasource.DriverManagerDataSource;
import org.springframework.orm.jpa.*;
import org.springframework.orm.jpa.vendor.HibernateJpaVendorAdapter;
import org.springframework.transaction.annotation.EnableTransactionManagement;
import org.springframework.web.servlet.config.annotation.*;

import javax.sql.DataSource;
import java.util.Properties;

@Configuration
@EnableWebMvc
@EnableTransactionManagement
@ComponentScan("com.example.course")
public class AppConfig implements WebMvcConfigurer {

   @Override
   public void configureDefaultServletHandling(DefaultServletHandlerConfigurer configurer)
{
      configurer.enable();
   }

    @Bean
   public DataSource dataSource() {
      DriverManagerDataSource ds = new DriverManagerDataSource();
      ds.setDriverClassName("com.mysql.cj.jdbc.Driver");
      ds.setUrl("jdbc:mysql://localhost:3306/studentdb?createDatabaseIfNotExist=true");
      ds.setUsername("root");
      ds.setPassword("password");
```

```
        return ds;
    }

    @Bean
    public LocalContainerEntityManagerFactoryBean entityManagerFactory(DataSource ds) {
        LocalContainerEntityManagerFactoryBean factory = new
LocalContainerEntityManagerFactoryBean();
        factory.setDataSource(ds);
        factory.setPackagesToScan("com.example.course");
        factory.setJpaVendorAdapter(new HibernateJpaVendorAdapter());

        Properties props = new Properties();
        props.put("hibernate.dialect", "org.hibernate.dialect.MySQLDialect");
        props.put("hibernate.hbm2ddl.auto", "update");
        props.put("hibernate.show_sql", "true");
        factory.setJpaProperties(props);

        return factory;
    }

 @Bean
Public JpaTransactionManager
transactionManager(LocalContainerEntityManagerFactoryBean factory) {
        return new JpaTransactionManager(factory.getObject());
    }
}
```